

Touch-Tone Dialer

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Presented here is a simple touch-tone dialer made using tone encoder TCM5087. It shows how dial tones are generated in telephone dialing systems.

The dialer operates with two groups of tones—the high-band tones and the low-band tones. The system requires one tone from each group to be generated each time a keypad push-button is pressed. These tones can be heard through a loudspeaker. The system can be used to dial any number wirelessly to demonstrate the concept of dialing.



Circuit and working

Fig. 1 shows circuit diagram of the touch-tone dialer. The dialer comprises tone encoder IC TCM5087 (IC1), which is designed especially for telephone use. Selection of the desired pair of tones is accomplished by the keypad comprising switches S1 through S12. Each switch shorts out one of the four rows (ROW1, ROW2, ROW3 and ROW4) to one of the three columns (COL1, COL2 and COL3).

Each divider section of IC1 is dedicated to generate one of the desired frequencies. High-band tones are selected via columns and low-band tones

via rows. A corresponding mix of both the selected tones is produced at pin 16 of IC1. The output is amplified by transistor T1, whose collector is connected

TABLE I
Tone Selection

Tone selection pins	Encoder output (Hz)
ROW1	701.3
ROW2	771.4
ROW3	857.2
ROW4	935.1
COL1	1215.9
COL2	1331.7
COL3	1471.9
COL4	1645

TABLE II
Test Points

Test point	Details
TP0	0V
TP1	9V
TP2	Frequency (tone output) depends on the pressed switch

PARTS LIST

Semiconductors:

- IC1 - TCM5087 tone dialer
- T1 - BC547 npn transistor

Resistor (1/4-watt, $\pm 5\%$ carbon):

- R1 - 220-ohm

Miscellaneous:

- S1-S12 - Push-to-on switch
- S13 - On/off switch
- LS1 - 8-ohm, 0.5W speaker
- BATT.1 - 9V PP3 battery
- X₁ - 3.579MHz crystal
- 16-pin IC base

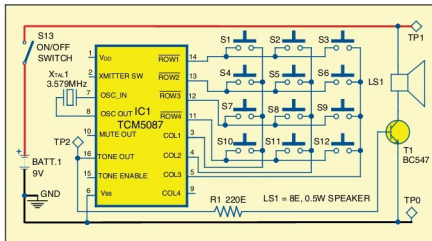


Fig. 1: Circuit diagram of the touch-tone dialer

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